Spaceability of Smooth Disk-Algebra Functions with Boundary Pringsheim Singularities

Math discovery smoke-test packet

June 12, 2026

1 Source

The source paper is

Andreas Debrouwere, *Some results about spaceability in function spaces*, arXiv:2606.13195.

Remark 3.15 points to the following question from Bernal-Gonzalez et al.:

$\mathbb{N}$. $\square$

Remark 3.15. Corollary 3.14 sheds some light on the question on [6, p. 607], which asked whether the smaller set

$$\{f \in A^\infty(\mathbb{D}) \mid \widetilde{f_{|\mathbb{T}}} \text{ is Pringsheim singular at every point of } \mathbb{R}\}$$

is spaceable in $A^\infty(\mathbb{D})$.

Here $A^\infty(\mathbb{D})$ denotes the smooth disk algebra. A boundary point is Pringsheim singular for the boundary restriction when the holomorphic function has no analytic continuation through that point. Thus it is enough to construct a closed infinite-dimensional subspace of $A^\infty(\mathbb{D})$ whose nonzero elements all have $\mathbb{T}$ as a natural boundary.

2 Claim

Theorem 1. *The set*

$$\{f \in A^\infty(\mathbb{D}) : f|_{\mathbb{T}} \textit{ is Pringsheim singular at every boundary point}\}$$

is spaceable in $A^\infty(\mathbb{D})$.

3 Proof

We use the classical Hadamard gap theorem in the following form.

Lemma 1 (Hadamard gap theorem)**.** *Let*

$$F(z) = \sum_{\ell=1}^{\infty} a_\ell z^{m_\ell}$$

have radius of convergence 1*, where* $m_{\ell+1}/m_\ell \geq q > 1$*. Then the unit circle is a natural boundary for* F*.*

Let the master lacunary sequence be $N_k = 2^k$, $k \geq 1$. Partition $\mathbb{N}$ into infinitely many infinite sets

$$A_m = \{2^m(2j+1) : j = 0, 1, 2, \ldots\}, \qquad m = 0, 1, 2, \ldots.$$

For $m \geq 0$, define

$$n_{m,j} = 2^{2^m(2j+1)}$$

and

$$g_m(z) = \sum_{j=0}^{\infty} \exp(-\sqrt{n_{m,j}})\, z^{n_{m,j}}.$$

The sets $\{n_{m,j} : j \geq 0\}$ are pairwise disjoint and all lie in the master support $\{2^k : k \geq 1\}$. Moreover,

$$\sum_j n_{m,j}^r \exp(-\sqrt{n_{m,j}}) < \infty$$

for every $r \geq 0$. Hence each derivative series of g_m converges uniformly on $\overline{\mathbb{D}}$, so $g_m \in A^\infty(\mathbb{D})$.

Let

$$X = \overline{\operatorname{span}\{g_m : m = 0, 1, 2, \ldots\}}^{A^\infty(\mathbb{D})}.$$

This is a closed infinite-dimensional subspace, since the g_m's have pairwise disjoint nonzero Taylor supports and are therefore linearly independent.

We claim that every nonzero $f \in X$ has $\mathbb{T}$ as a natural boundary. Write

$$f(z) = \sum_{n=0}^{\infty} c_n z^n.$$

The coefficient functionals $f \mapsto c_n = f^{(n)}(0)/n!$ are continuous on $A^\infty(\mathbb{D})$. Since finite linear combinations of the g_m's have no coefficients outside the master set, every $f \in X$ also satisfies

$$c_n = 0 \qquad (n \notin \{2^k : k \geq 1\}).$$

Likewise, for fixed m, finite linear combinations satisfy

$$\frac{c_{n_{m,j}}}{\exp(-\sqrt{n_{m,j}})} = \frac{c_{n_{m,j'}}}{\exp(-\sqrt{n_{m,j'}})} \qquad (j, j' \geq 0),$$

and this equality is preserved under closure. Therefore each $f \in X$ has scalars λ_m such that

$$c_{n_{m,j}} = \lambda_m \exp(-\sqrt{n_{m,j}}) \qquad (j \geq 0).$$

If $f \neq 0$, then $\lambda_m \neq 0$ for at least one m; otherwise all Taylor coefficients of f would be zero.

For such an m,

$$\lim_{j\to\infty} |c_{n_{m,j}}|^{1/n_{m,j}} = \lim_{j\to\infty} |\lambda_m|^{1/n_{m,j}} \exp(-1/\sqrt{n_{m,j}}) = 1.$$

Since $f \in A^\infty(\mathbb{D})$, its radius of convergence is at least 1. The preceding limit shows that the radius is exactly 1. The nonzero Taylor support of f is a subsequence of the master support $\{2^k\}$; therefore, when listed increasingly, its successive exponents have ratio at least 2. Hadamard's gap theorem applies and gives $\mathbb{T}$ as a natural boundary for f. Thus $f|_{\mathbb{T}}$ is Pringsheim singular at every boundary point.

Therefore $X \setminus \{0\}$ is contained in the set in the theorem, and the set is spaceable. $\square$

4 Verification Notes

The script `code/check_lacunary_construction.py` checks finite samples of the partition and coefficient profile:

```
conda run --no-capture-output -n sandbox python code/check_lacunary_construction.py
```

The run confirmed disjoint blocks, a master consecutive ratio at least 2, and the expected root behavior of $\exp(-\sqrt{n})$. These checks are only sanity checks; the proof above is analytic and uses Hadamard's theorem.

5 Limitations and Review Focus

The main external ingredient is the classical Hadamard gap theorem. A human reviewer should verify that the terminology "Pringsheim singular at every point" in Bernal-Gonzalez et al. agrees with the natural-boundary conclusion used here. The construction gives the stronger conclusion that no nonzero element of X analytically continues through any boundary point.

Human Review: Pringsheim-singular spaceability question

Original automatic proof: `solution_packet.pdf`
Checked date: 13 June 2026
Status: Manually checked.
Verdict: Not a solution to the original question as stated in Bernal-Gonzalez–Bonilla–Lopez-Salazar–Seoane-Sepulveda.

Human Comments

The proposed construction appears to prove a weaker natural-boundary statement, but it does not prove the Pringsheim-singularity condition required in the original question.

The source packet interprets "Pringsheim singular at every boundary point" as meaning that the holomorphic function has no analytic continuation through any point of $\mathbb{T}$. Under this interpretation, Hadamard's gap theorem is a natural tool: the constructed functions have lacunary Taylor support and radius of convergence exactly 1, so the unit circle is a natural boundary.

However, the original paper

Bernal-Gonzalez–Bonilla–Lopez-Salazar–Seoane-Sepulveda,

Nowhere holderian functions and Pringsheim singular functions in the disc algebra, defines Pringsheim singularity for the boundary function in a stronger real-variable sense. If

$$f_0(t) = f(e^{it}),$$

then $f|_{\mathbb{T}}$ is Pringsheim-singular at e^{it_0} when the Taylor series of f_0 at t_0 has radius of convergence zero. Equivalently,

$$\limsup_{r\to\infty} \left| \frac{f_0^{(r)}(t_0)}{r!} \right|^{1/r} = +\infty.$$

This is stronger than merely saying that f has no holomorphic continuation through e^{it_0}.

Where the Proof Falls Short

The packet proves that every nonzero element of the constructed closed span has $\mathbb{T}$ as a natural boundary. This rules out analytic continuation through any boundary point, and plausibly gives a nowhere-real-analytic boundary function. But it does not establish the derivative-growth condition above.

For a boundary function

$$f_0(t) = \sum_{n\in\Lambda} c_n e^{int},$$

one has

$$f_0^{(r)}(t_0) = \sum_{n\in\Lambda} (in)^r c_n e^{int_0}.$$

To prove Pringsheim singularity in the sense of the original paper, one would need pointwise lower bounds on these derivative sums at every t_0, strong enough to force

$$\left| f_0^{(r)}(t_0)/r! \right|^{1/r}$$

to be unbounded.

The coefficient choice $c_n = e^{-\sqrt{n}}$ is suggestive: a single term of the form $n^r e^{-\sqrt{n}}$ can be very large compared with $r!$ for appropriate n. But the proof does not show that such a term dominates the whole derivative sum, nor does it rule out cancellations. This is especially important for general nonzero elements of the closed span, where several lacunary blocks can contribute simultaneously with complex coefficients.

Review Outcome

This packet should be rejected as a solution to the original Pringsheim spaceability question. It should not be archived as a correct full answer to the question in [6].

It may still be useful as a proof of a weaker statement: spaceability of smooth disk-algebra functions whose unit circle is a natural boundary, or possibly whose boundary restrictions are nowhere real analytic. But the argument does not prove that the boundary Taylor radius is zero at every point, which is the condition required for Pringsheim singularity in the original reference.